

Week 06: Swarm Intelligence: Particles, Ants and Bees

Artificial Intelligence Applications in Engineering (MUH-920 Mühendislikte Yapay Zeka Uygulamaları). Lecture notes

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Week overview, study path and tasks: <https://github.com/utkukose/ai-applications-in-engineering-NB-lecture/blob/main/weeks/week-06/README.md>

Lecture page with animations and knowledge checks: <https://utkukose.github.io/ai-applications-in-engineering-NB-lecture/weeks/week-06/lecture.html>

Interactive lab with self-assessment and reflection: <https://utkukose.github.io/ai-applications-in-engineering-NB-lecture/weeks/week-06/lab.html>

Colab notebook with Python step 6 and the hands-on work:

https://colab.research.google.com/github/utkukose/ai-applications-in-engineering-NB-lecture/blob/main/weeks/week-06/NB06_swarm_intelligence.ipynb

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Overview

Flocks, ant colonies and bee hives solve problems that no individual could solve alone, using simple local rules and indirect communication. Swarm intelligence turns these mechanisms into optimisation algorithms [1]. Particle swarm optimisation moves candidate solutions through a continuous space under the pull of their own best and the swarm's best positions [2], ant colony optimisation builds tours with pheromone trails [3, 4], and the artificial bee colony algorithm divides the search among employed, onlooker and scout bees [5]. This week tunes a PID speed controller of a DC motor with a particle swarm, plans a drilling sequence for a circuit board with ants, and asks how to judge the many nature-inspired algorithms critically [6].

Estimated study time: 9 to 11 hours.

Midterm project. The midterm project is released this week and is due at the end of Week 7. It applies search, knowledge-based and fuzzy systems, and evolutionary and swarm optimisation to a problem of the student's department, and it counts 40 percent of the course grade. The [project page](#) describes the three options, the starter notebooks, the deliverables and the evaluation criteria.

Learning outcomes

By the end of the week, students are expected to explain the velocity and position update of particle swarm optimisation and the role of inertia, cognitive and social coefficients, to implement a particle swarm as a Python class, to formulate controller tuning as an optimisation problem with a time-domain cost, to explain how pheromone deposit and evaporation guide ant colony optimisation, to apply it to a sequencing problem, and to evaluate claims about new metaheuristics with sound experimental practice.

Python in this week

These notes close with Python step 6: Classes, objects and probabilistic choice. It covers classes, objects, attributes and methods for a DC motor model and the particles of a swarm, and the probabilistic choice and pheromone update of ant colony optimisation with NumPy [3, 7]. In the course materials, the step is taught at the start of the Colab notebook, where each snippet runs in a cell next to its explanation and short exercises give immediate feedback, and later sections of the notebook mark further constructs as Python moves.

Collective behaviour as computation

A flock of birds turns as one body although no bird leads it. Reynolds reproduced this with three local rules for each simulated bird: avoid crowding neighbours, align with their heading, and move towards their centre [8]. Ant colonies find short paths between nest and food without a map, because ants deposit pheromone and prefer trails with more of it; shorter trails are completed faster and accumulate pheromone sooner. Honeybee colonies allocate foragers to flower patches in proportion to their quality through the waggle dance. These systems share decentralised control, simple agents, local interaction and emergent global behaviour, which Bonabeau, Dorigo and Theraulaz summarised as swarm intelligence [1].

Animation: Three rules make a flock. Each arrow follows only its neighbours. Set separation, alignment and cohesion and watch order emerge or dissolve. With alignment at zero, the flock never forms a common heading. Run it on the [lecture page](#).

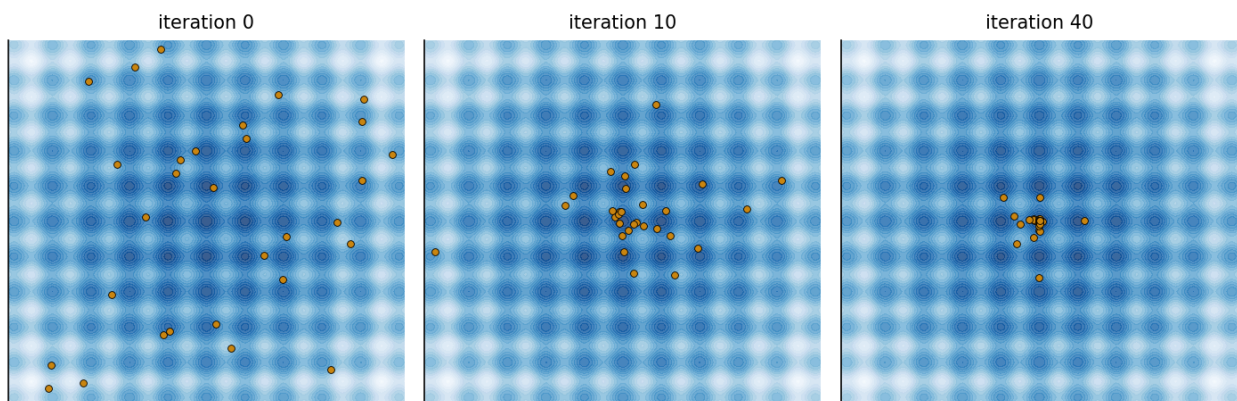


Figure 6.1. Particle swarm optimisation on a landscape with many local minima: positions of the swarm after 0, 10 and 40 iterations. The swarm first explores and then contracts around the best region [2].

Particle swarm optimisation

Kennedy and Eberhart introduced particle swarm optimisation in 1995, inspired by bird flocking and social behaviour [2]. Each particle i has a position x , which is a candidate solution, and a velocity v . It remembers the best position p it has visited, and the swarm shares the best position g found by any particle. In every iteration, the velocity is updated as $v = w v + c_1 r_1 (p - x) + c_2 r_2 (g - x)$, and the position as $x = x + v$, where r_1 and r_2 are random numbers between 0 and 1 drawn for each dimension.

The three terms have clear meanings. The inertia weight w keeps part of the previous motion; Shi and Eberhart introduced it to balance global and local search, with values around 0.4 to 0.9 [9]. The cognitive term, weighted by c_1 , pulls a particle back towards its own best experience. The social term, weighted by c_2 , pulls it towards the best experience of the swarm. Clerc and Kennedy analysed the dynamics and derived a constriction factor that guarantees convergence; the widely used setting $w = 0.7298$ with $c_1 = c_2 = 1.49618$ follows from it [10]. Particle swarms are easy to implement, need no gradients, and work well on continuous problems of moderate dimension, but like every metaheuristic they can stagnate in a local optimum.

Check your understanding. In the velocity update, what does the social term $c_2 r_2 (g - x)$ do?

- A. It slows every particle down
- B. It pulls the particle towards the best position found by the whole swarm
- C. It pulls the particle towards its own best position
- D. It adds random noise independent of any position

Answer: B. g is the global best position shared by the swarm, so this term transmits the swarm's experience to each particle.

Tuning a PID controller with a swarm

The proportional-integral-derivative controller is the workhorse of industrial control. Its three gains shape the response: The proportional gain speeds up the reaction, the integral gain removes steady-state error, and the derivative gain damps oscillation [11]. Tuning them by hand is tedious, and classical rules assume simple process models. Controller tuning can instead be formulated as optimisation: Choose the gains that minimise a cost computed from a simulated step response, such as the integral of the time-weighted absolute error (ITAE), with penalties for overshoot and limits on the actuator. Particle swarms have been used in this way for automatic voltage regulators and many other loops [12].

The notebook applies the method to the speed control of a DC motor, using the standard model and parameters of the Control Tutorials for MATLAB and Simulink [13]. The armature voltage is limited to 24 volts, which makes the problem nonlinear, and the swarm evaluates all particles in one vectorised simulation. The cost function is part of the design: A cost that rewards speed alone drives the gains to their bounds, and adding a penalty on overshoot or control effort changes the optimum. The optimiser serves the engineer's definition of good behaviour; it cannot supply that definition.

Check your understanding. A swarm tunes a PID controller by minimising only the settling time, and the result shows violent oscillation of the actuator. What is the most likely cause?

- A. The swarm has too many particles
- B. The cost function does not penalise control effort or oscillation, so the optimiser exploits this
- C. PID controllers cannot be tuned by optimisation
- D. The inertia weight is exactly 0.7298

Answer: B. An optimiser minimises exactly what the cost states; unpenalised behaviour is free for it to exploit.

Ant colony optimisation

Dorigo, Maniezzo and Colorni turned the pheromone mechanism into the Ant System for the travelling salesman problem [3]. Each artificial ant builds a complete tour city by city. From city i , it chooses the next unvisited city j with a probability proportional to the pheromone on edge (i, j) raised to a power α , times a heuristic desirability, usually the inverse distance, raised to a power β . After all ants have finished, pheromone evaporates on every edge by a factor ρ , and each ant deposits pheromone on the edges of its tour in inverse proportion to the tour length. Short tours therefore reinforce their edges, while evaporation forgets poor early choices.

The Ant Colony System added a stronger exploitation rule and local pheromone updates and was competitive on benchmark instances [14]; the monograph by Dorigo and Stützle presents the family and its applications to routing, scheduling and assignment [4]. Drilling holes in a printed circuit board is a classical travelling salesman application: The drill head must visit every hole once, and the travel time between holes is wasted production time. Benchmark libraries such as TSPLIB include instances that come from drilling problems [15].

Check your understanding. What is the purpose of pheromone evaporation in ant colony optimisation?

- A. It makes all tours equally likely forever
- B. It gradually removes pheromone from edges that are not reinforced, so poor early choices are forgotten

- C. It increases the length of short tours
- D. It fixes the start city

Answer: B. Without evaporation, early random choices would accumulate pheromone and lock the colony into poor tours.

Artificial bee colony and the metaphor question

Karaboga and Basturk's artificial bee colony algorithm assigns roles to solutions [5]. Employed bees search around their food sources, onlooker bees choose sources in proportion to their quality and search around them, and scout bees replace sources that have not improved for a number of trials with random new ones. The scout phase gives the algorithm a built-in restart mechanism. Many further algorithms followed, inspired by fireflies, bats, wolves, whales and others, including the ant lion optimiser used to train neural networks for chaotic signal prediction in earlier work [16].

Sörensen argued that the flood of metaphor-based algorithms often hides old ideas behind new vocabulary and weak experiments [6]. The no free lunch theorems add that no algorithm can be best on all problems [17]. Engineers should therefore judge a metaheuristic by its search mechanism rather than its metaphor, compare it with established methods on the problem at hand, report results over many independent runs with statistics, and count function evaluations, not iterations, because a single evaluation of an engineering simulation may take minutes.

Check your understanding. A paper claims that a new animal-inspired algorithm beats PSO, based on one run per method with different numbers of function evaluations. What is the main weakness?

- A. The metaphor is not biological enough
- B. One run and unequal evaluation budgets do not support a fair comparison of stochastic algorithms
- C. PSO cannot be compared with other algorithms
- D. The paper should have used a larger swarm only for the new algorithm

Answer: B. Fair comparisons need equal budgets, many independent runs and statistical summaries.

Python step 6: Classes, objects and probabilistic choice

Why classes

Simulation code keeps state that changes over time, such as the speed and the current of a motor, together with the functions that change it. A class bundles both: Attributes store the state, and methods are functions that act on it. The class is the blueprint, and every object created from it has its own attributes. The special method `__init__` runs when an object is created and sets its initial state, and the first parameter of every method, `self`, refers to the object itself.

The DC motor of this week has the parameters of the speed example of the Control Tutorials for MATLAB and Simulink [13]. Its speed w and current i obey $J dw/dt = K i - b w$ and $L di/dt = V - R i - K w$, which the method `step` integrates over small time steps.

```
class DCMotor:
    """Armature-controlled DC motor with the parameters of the CTMS
    speed example."""

    def __init__(self, J=0.01, b=0.1, K=0.01, R=1.0, L=0.5):
        self.J, self.b, self.K, self.R, self.L = J, b, K, R, L
        self.speed = 0.0          # rad/s
        self.current = 0.0        # A

    def step(self, voltage, dt=0.001):
        """Advance the motor by dt seconds with the explicit Euler
        method."""
        acc = (self.K * self.current - self.b * self.speed) / self.J
```

```

        di = (voltage - self.R * self.current - self.K * self.speed) /
self.L
        self.speed += acc * dt
        self.current += di * dt
        return self.speed

```

```

motor = DCMotor()
for _ in range(3000):          # three seconds in steps of 1 ms
    motor.step(1.0)
print(f"speed after 3 s at 1 V: {motor.speed:.4f} rad/s")

```

Output

speed after 3 s at 1 V: 0.0996 rad/s

The underscore `_` is the conventional name for a loop variable that is not used. The speed approaches $K / (bR + K^2)$, about 0.1 rad/s for 1 V, which the steady-state analysis of the tutorial also gives.

Many objects from one class

Objects created from the same class are independent. A motor with a lighter rotor does not share its speed with a motor with a heavier one, and each object keeps its own attributes. This independence makes classes practical for swarms, fleets of vehicles and banks of sensors.

```

light = DCMotor(J=0.005)
heavy = DCMotor(J=0.05)
for _ in range(200):          # 0.2 s
    light.step(1.0)
    heavy.step(1.0)
print(f"light rotor: {light.speed:.4f} rad/s, heavy rotor:
{heavy.speed:.4f} rad/s")

```

Output

light rotor: 0.0257 rad/s, heavy rotor: 0.0061 rad/s

A particle of the swarm

Particle swarm optimisation moves each particle with a velocity that combines its previous velocity, a pull towards its own best position and a pull towards the best position of the swarm. A class keeps the position, the velocity and the personal best of a particle together, and NumPy arrays let the same code work in any number of dimensions. The method `copy` creates an independent array: Without it, the personal best and the current position would be two names for the same array, and changing one in place would also change the other.

```
import numpy as np
```

```

class Particle:
    """One particle of a swarm with position, velocity and personal
    best."""

    def __init__(self, position, velocity):
        self.x = np.array(position, dtype=float)
        self.v = np.array(velocity, dtype=float)
        self.best_x = self.x.copy()

    def move(self, swarm_best, w=0.7, c1=1.5, c2=1.5, r1=0.5, r2=0.5):

```

```

        """Update velocity and position; r1 and r2 stand in for random
        numbers."""
        self.v = w * self.v + c1 * r1 * (self.best_x - self.x) + c2 *
r2 * (swarm_best - self.x)
        self.x = self.x + self.v

```

```

p = Particle([2.0, -1.0], [0.1, 0.0])
p.move(swarm_best=np.array([0.0, 0.0]))
print(p.x, p.v)

```

Output

```
[ 0.57 -0.25] [-1.43  0.75]
```

The notebook of this week builds a complete swarm on the same pattern, with new random numbers at every step.

Probabilistic choice in an ant colony

Ant colony optimisation builds solutions step by step [3]. At every step an ant chooses among the possible next moves with probabilities proportional to $\tau^\alpha \eta^\beta$, where τ is the pheromone on a move and η its heuristic desirability, usually one over the distance. The generator method `choice` draws from such a discrete distribution when the probabilities are passed as `p`, and `np.bincount` counts how often each move was chosen.

```

tau = np.ones(4)                                # pheromone on four
candidate edges
dist = np.array([4.0, 2.0, 5.0, 3.0])           # edge lengths in km
alpha, beta = 1.0, 2.0
weight = tau ** alpha * (1 / dist) ** beta
prob = weight / weight.sum()
print(prob.round(3))
colony = np.random.default_rng(5)
choices = colony.choice(4, size=1000, p=prob)    # the choices of 1000
ants
print(np.bincount(choices, minlength=4))

```

Output

```
[0.135 0.539 0.086 0.24 ]
[148 532  78 242]
```

After the ants have moved, pheromone evaporates at a rate ρ , and every ant deposits pheromone on its choice, more on shorter edges. Repeated over many iterations, this positive feedback concentrates the colony on short routes, while evaporation keeps it from settling too early on a poor one.

```

rho = 0.5
deposit = np.bincount(choices, minlength=4) / dist / len(choices)  #
1/length per ant, scaled by the colony size
tau = (1 - rho) * tau + deposit
print(tau.round(3))

```

Output

```
[0.537 0.766 0.516 0.581]
```

Check your understanding. A class `Counter` sets `self.n = 0` in `init` and has a method `add` that runs `self.n += 1`. After `a = Counter()`, `b = Counter()`, `a.add()`, `a.add()` and `b.add()`, what are `a.n` and `b.n`?

- A. 2 and 1
- B. 3 and 3
- C. 1 and 1
- D. 2 and 2

Answer: A. Each object keeps its own attributes, so the calls on a do not change b.

Discipline challenges

Swarms tune continuous parameters and ants build sequences. Choose a row, write the cost function, and compare the swarm with a simple baseline.

Department	Challenge
Electrical and Electronics Engineering	PID tuning of a DC motor or of a buck converter output voltage loop, as in the notebook.
Mechanical Engineering and Automotive Engineering	Suspension damping and stiffness that trade comfort against road holding.
Industrial Engineering	Vehicle routing for deliveries from a depot with ant colony optimisation.
Computer Engineering	Routing in a communication network or task allocation among cores with ants.
Civil Engineering	Size optimisation of a planar truss with stress constraints using PSO.
Environmental Engineering	Calibration of a river water-quality model to measured oxygen profiles.
Mining Engineering	Sequencing of blast holes or of stope extraction with ants.
Geophysical Engineering	Inversion of a layered-earth resistivity model from sounding curves with PSO.
Chemical Engineering and Chemistry	Estimation of kinetic parameters of a reaction from concentration curves.
Textile Engineering	Cutting-order planning and marker layout sequencing.
Biology and Statistics	Parameter estimation of a population growth model, compared with least squares.

Weekly task

Apply particle swarm optimisation or ant colony optimisation to a problem from your department. For a continuous problem, compare PSO with differential evolution from Week 5 under the same evaluation budget over ten seeds. For a sequencing problem, compare the ant colony with the nearest-neighbour heuristic. Report the setup, a convergence plot and a statistical summary in about 500 words, citing at least three works from this week's references [2, 3, 6].

Research and report assignment (optional)

Swarm intelligence in engineering practice. Review applications of particle swarm, ant colony or bee colony algorithms in one field, such as power systems, structural design, logistics or controller tuning. Assess the quality of the experimental comparisons in the reviewed papers against the criteria of Sørensen and against the no free lunch theorems [6, 12, 17].

Both assignments are optional. During an active semester, the weekly task can be sent together with the learning log exported from the interactive lab, and the research report on its own, to utkukose@sdu.edu.tr or utkukose@gmail.com. They carry no separate weight, but the instructor may

take them into account as a discretionary adjustment of the midterm component of the grade. The assessment section of the syllabus gives the details, and research reports follow the report template of the course repository.

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